

eSDK Enterprise Storage CSM Plugins V2.7.0

User Guide

Issue	01
Date	2026-06-30



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About This Document

Intended Audience

This document is intended for:

- Technical support engineers
- O&M engineers
- Engineers with basic knowledge of storage, Kubernetes, and CSI

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
 NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
 NOTE	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Change History

Issue	Date	Description
01	2026-03-30	This issue is the first official release.

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1 Product Description

Container Storage Monitor (CSM) is a tool used for visual display of Huawei storage resources and Kubernetes resources in Kubernetes container scenarios. This tool can notify storage of the relationship between a PV/Pod and a LUN/file system so that the relationship can be displayed on the storage for storage administrators to view. It can also upload the performance, capacity, IOPS, and other data of a LUN/file system to a third-party network management system for application administrators to view. In this way, O&M availability in container scenarios can be improved.

2 Compatibility and Features

- [2.1 Kubernetes Compatibility](#)
- [2.2 Compatibility with Huawei Storage](#)
- [2.3 csm-prometheus Feature Description](#)
- [2.4 csm-storage Feature Description](#)
- [2.5 Constraints](#)

2.1 Kubernetes Compatibility

Table 2-1 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16~1.35
Red Hat OpenShift Container Platform	4.12~ 4.20
CCE Agile	22.3.2

2.2 Compatibility with Huawei Storage

NOTE

- When CSM is used to interconnect with storage, only LUNs/file systems provisioned by CSI can be displayed. Dtrees are not supported.
- Only OceanStor 6.1.7 and later and OceanStor Dorado 6.1.7 and later support the CSM-Storage label function.

Table 2-2 CSM-Storage compatibility

Storage Version	Huawei CSI Version
OceanStor 6.1.7/6.1.8/ V700R001C00/V700R001C10/ V700R001C20 OceanStor Dorado 6.1.7/6.1.8/ V700R001C00/V700R001C10/ V700R001C20	4.12.0

Table 2-3 CSM-Prometheus compatibility

Storage Version	Prometheus Version	Grafana Version	Huawei CSI Version
OceanStor 6.1.3/6.1.5/6.1.6/6.1.7/6.1.8/ V700R001C00/ V700R001C10/ V700R001C20 OceanStor Dorado 6.1.0 ¹ /6.1.2/6.1.3/6.1.5/ 6.1.6/6.1.7/6.1.8/ V700R001C00/ V700R001C10/ V700R001C20	2.25.0 - 2.43.0	Grafana 11.6.1 - 12.1.1	4.12.0

- Note 1: If OceanStor Dorado 6.1.0 is used, the display of Prometheus performance monitoring metrics may be broken or discontinuous as the number of concurrent requests increases.

2.3 csm-prometheus Feature Description

csm-prometheus collects storage monitoring data and exposes the data to the Prometheus platform for collection. The following table lists the supported monitoring metrics.

Table 2-4 Object metrics supported by monitored objects of centralized storage

Object Type	Object Metric
Storage	• Basic information • Monitoring status • Running status

Object Type	Object Metric
Controller	• CPU usage • Memory usage • Health status • Running status
Storage pool	• Total capacity • Remaining capacity • Used capacity • Capacity usage
LUN	• Total capacity • Capacity usage
File system	• Total capacity • Capacity usage • Used snapshot capacity
PV	• Total capacity • Capacity usage
vStore	• Total quota • Remaining quota • Used quota • Quota usage

Table 2-5 Performance metrics supported by monitored objects of centralized storage

Object Type	Performance Metric
Controller	• 21: bandwidth (MB/s) • 23: read bandwidth (MB/s) • 26: write bandwidth (MB/s) • 22: IOPS • 25: read IOPS • 28: write IOPS • 370: average I/O response time (μs)
Storage pool	• 21: bandwidth (MB/s) • 22: IOPS • 370: average I/O response time (μs)

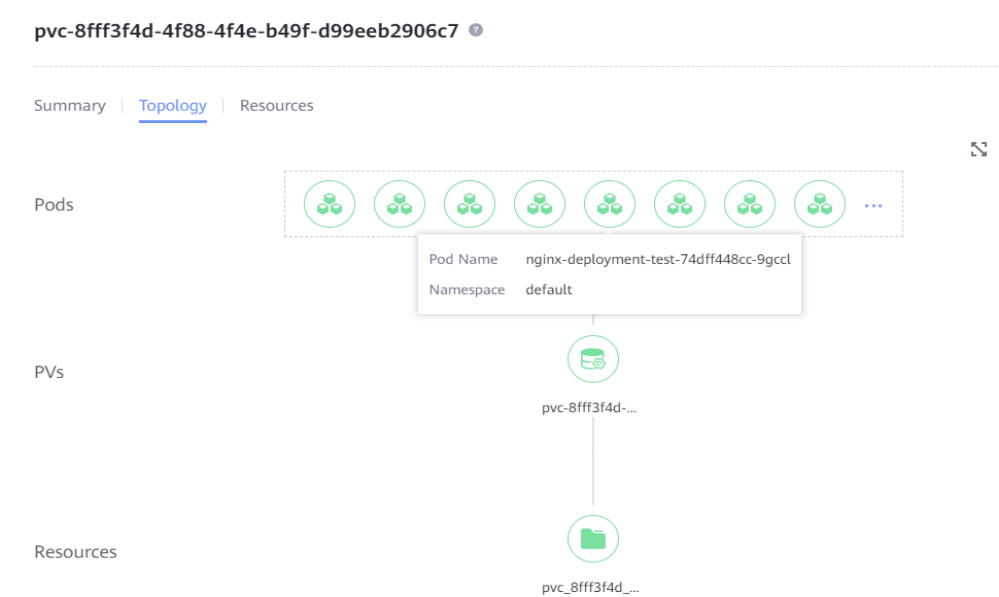
Object Type	Performance Metric
LUN	• 21: bandwidth (MB/s) • 22: IOPS • 370: average I/O response time (μs)
File system	• 182: OPS • 524: average read OPS response time (μs) • 525: average write OPS response time (μs)
PV	• LUN bandwidth: bandwidth (MB/s) of a PV of the LUN type on the storage • LUN IOPS: IOPS of a PV of the LUN type on the storage • Average LUN I/O response time: average I/O response time (μs) of a PV of the LUN type on the storage • File system OPS: OPS of a PV of the file system type on the storage • Average file system read OPS response time: average read OPS response time (μs) of a PV of the file system type on the storage • Average file system write OPS response time: average write OPS response time (μs) of a PV of the file system type on the storage

2.4 csm-storage Feature Description

NOTICE

The default memory quota of the CSM topology service is 512 Mi. The memory usage of the topology service increases linearly with the number of PVs and Pods in the cluster. If there are a large number of resources in the cluster, you can manually modify the memory quota of the topology service to ensure that the csm-storage feature can be used properly. For configuration details, see [Table 3-7](#).

csm-storage reports the PVs and the topology relationships with associated Pods, file systems, and LUNs to the storage system. The following figure shows the resource display effect on the storage GUI.



2.5 Constraints

Performance Constraints

Table 2-6 CSM specifications management

Specifications Management		Recommended Value
Storage management	Number of connected storage devices	5
	Number of CSMs interconnected with a single storage device	≤ 3
Monitoring item management	Maximum number of monitoring items supported by a storage device	15000
	Maximum number of monitoring items of all storage devices	40000
Scraping interval management	Object data scraping interval (for metrics in Table 2-4)	300s
	Performance data scraping interval (for metrics in Table 2-5)	300s

NOTICE

- If the data scraping interval is too small, the storage backend pressure will increase, adversely impacting the running of the storage backend.
- It is recommended that the object data scraping interval be different from the performance data scraping interval to prevent query failures caused by heavy storage backend pressure.
- A single storage resource (LUN or file system) does not support repeated topology creation.
- Topology relationships cannot be reported for resources with the same name of different vStores on the same storage device.

Table 2-7 Concurrent data scraping requests recommended by CSM

Total Number of Storage File Systems and LUNs	Recommended Number of Concurrent Requests
< 2000	20
2000-5000	10
> 5000	5

NOTICE

The default number of concurrent CSM data scraping requests is 20. When the number of storage resources increases, you are advised to configure the number of concurrent data scraping requests based on the recommended values to reduce the storage query load. For details about how to configure the number of concurrent requests, see [10.2 Manually Adjusting the Number of Concurrent Data Scraping Requests](#).

3 Installation Preparations

[3.1 Obtaining Tools and Software Packages](#)

[3.2 Uploading Images to the Image Repository](#)

[3.3 Installing Helm](#)

[3.4 Preparing the Configuration File](#)

3.1 Obtaining Tools and Software Packages

Tools

Table 3-1 lists the tools required for software installation, configuration, and commissioning.

Table 3-1 Required tools

Tool	Description	How to Obtain
PuTTY	Cross-platform remote access tool. It is used to access a node running a Windows OS during software installation.	You can visit the chiark homepage to download the PuTTY software. You are advised to use PuTTY of the latest version to ensure successful login to the storage system.
WinSCP	Cross-platform file transfer tool. Use version 5.7.5 or later and select SCP during file transfer. It is used to transfer files between Windows and Linux.	You can visit the WinSCP homepage to download the WinSCP software.

Software Packages

Before deploying services, you need to prepare the CSM software installation packages listed in [Table 3-2](#).

Table 3-2 Required software packages

Package	Description	How to Obtain
eSDK_Enterprise_Storage_CSM_V2.7.0_X86_64.zip eSDK_Enterprise_Storage_CSM_V2.7.0_ARM_64.zip	CSM software installation package.	https://github.com/Huawei/csm/releases

NOTE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file together with the software package for integrity verification.

After the software package is downloaded from Huawei Support website, verify its PGP digital signature by referring to *OpenPGP Signature Verification Guide*. If the verification fails, do not use the software package and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with.

- For carrier users, visit <https://support.huawei.com/carrier/digitalSignatureAction>.
- For enterprise users, visit <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>.

3.2 Uploading Images to the Image Repository

Prerequisites

- The required CSM software package is available. For details about how to obtain it, see [3.1 Obtaining Tools and Software Packages](#).
- An image repository has been prepared and can communicate with worker nodes. In addition, you have obtained the IP address, account, and password of the image repository.
- A project has been created in the image repository.
- A Linux host with Docker installed is available, and the host can access the image repository.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to the Linux host where Docker is installed through the management IP address.
- Step 2** Use WinSCP to upload the software package to the `/opt` directory.
- Step 3** Run the `unzip /opt/Software package name` command to decompress the software package.

Software package name indicates the CSM software package name.

 NOTE

The software package name **eSDK_Enterprise_Storage_CSM_V2.7.0_X86_64.zip** is used as an example.

```
# unzip /opt/eSDK_Enterprise_Storage_CSM_V2.7.0_X86_64.zip -d /opt/huawei-csm
```

Step 4 Run the following command and enter the account and password to log in to the image repository:

```
docker login <Image repository IP address>
```

Step 5 Run the following command to upload the CSM image.

```
chmod +x /opt/huawei-csm/helm/huawei-csm/upload-image.sh; ./opt/huawei-csm/helm/huawei-csm/  
upload-image.sh --imageRepo <Image repository project name>
```

----End

NOTICE

- The **upload-image.sh** script is interpreted and executed using the Bash (Bourne-Again Shell). Before executing the script, ensure that the current system supports the Unix shell of the Bash type.
 - For details about how to import and upload images to the CCE Agile platform, see the user manual of the platform.
-

3.3 Installing Helm

 NOTE

Helm 3 or later is supported.

Helm is a software package management tool in the Kubernetes ecosystem. Similar to Advanced Packaging Tool (APT) of Ubuntu, Yellowdog Updater, Modified (YUM) of CentOS, or Package Installer for Python (PIP) of Python, Helm manages Kubernetes application resources. You can use Helm to package, distribute, install, upgrade, and roll back Kubernetes applications in a unified manner.

- For details about how to obtain and install Helm, [click here](#).
- For other information about Helm, [click here](#).

3.4 Preparing the Configuration File

When using Helm, you need to prepare the **values.yaml** file based on the Huawei storage connected during deployment and the features to be used.

Procedure

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Use WinSCP to upload the software package to the **/opt** directory.

Step 3 Run the **unzip /opt/*Software package name*** command to decompress the software package.

Software package name indicates the software package name.

 NOTE

The software package name **eSDK_Enterprise_Storage_CSM_V2.7.0_X86_64.zip** is used as an example.

```
# unzip /opt/eSDK_Enterprise_Storage_CSM_V2.7.0_X86_64.zip -d /opt/huawei-csm
```

Step 4 Run the **cd /opt/huawei-csm/helm/huawei-csm** command to go to the Helm working directory.

Step 5 Run the **vi values.yaml** command to set parameters in the **values.yaml** file. After the modification is complete, press **Esc** and enter **:wq!** to save the modification. [Table 3-3](#), [Table 3-4](#), [Table 3-5](#), and [Table 3-6](#) describe related parameters.

The **global** configuration items are used to configure the global information required by the system.

Table 3-3 global configuration items

Parameter	Description	Mandatory	Default Value
replicaCount	Number of Pod copies corresponding to the Deployment deployed using CSM. When the number of Pod copies is greater than 1, the system automatically enables the leaderElection function.	No	1. It is recommended that the value be less than or equal to 2. • If the number of Pod copies is 1: Ensure that the number of CPU cores is 4 or above and the memory is 4 GB or above in Kubernetes. • If the number of Pod copies is 2: Ensure that the number of CPU cores is 8 or above and the memory is 8 GB or above in Kubernetes. NOTICE – If the single-copy mode is used during deployment and you want to change it to the multi-copy mode, update the replicaCount parameter by following the instructions in 4.1.3 Upgrading the Software. – Do not use other methods to change the number of Pod copies corresponding to the Deployment deployed using CSM. If you use other methods, for example, directly editing the Spec.replicas parameter in a Deployment deployed using CSM, the multi-copy function will be abnormal.
imageRepo	Image repository name.	Yes	None. The value must be the same as the image repository name in Step 5 .

Parameter	Description	Mandatory	Default Value
logging.module	Log mode.	Yes	file The value can be file or console . NOTICE If the permission is insufficient when the file mode is used (for example, in the OpenShift environment), manually plan the log directory by following the instructions in 10.4 The Pod Status Is CrashLoopBackOff and the Log Contains "mkdir permission denied" .
logging.level	Log level.	Yes	info The value can be debug , info , warning , or error .
logging.fileSize	Log file size.	Yes	20 MB
logging.maxBackups	Maximum number of backup logs.	Yes	9
leaderElection.leaseDuration	Leader duration. This parameter takes effect only in the multi-copy mode.	No	8s
leaderElection.renewDeadline	Time for the leader to be re-elected. This parameter takes effect only in the multi-copy mode.	No	6s
leaderElection.retryPeriod	Leader election retry time. This parameter takes effect only in the multi-copy mode.	No	2s
balancedDeploy	If this parameter is set to true , Pods of different services will be scheduled to different nodes during CSM installation.	No	true If the schedule of balancedDeploy conflicts with that of nodeSelector , the schedule of balancedDeploy does not take effect.

Parameter	Description	Mandatory	Default Value
pullPolicy	Image pull policy.	Yes	IfNotPresent
imagePullSecrets	Used by the Kubernetes cluster to pass the identity authentication of an image registry to pull private images.	No	None. For details, see Pull an Image from a Private Registry .
namespace	Namespace where the CSM plug-in is deployed.	Yes	huawei-csm
csiDriverNamespace	Namespace where the CSI driver is located.	Yes	huawei-csi

The **features** configuration items are used to enable and disable features.

Table 3-4 features configuration items

Parameter	Description	Mandatory	Default Value	Remarks
prometheusCollector.enabled	Whether to enable the Prometheus collection service.	Yes	true	-
prometheusCollector.nodePort	Prometheus nodePort.	No	30074	The default value is 30074 . This port is a host port. If a conflict occurs, change the value.
prometheusCollector.csiDriverName	Registered CSI driver name.	Yes	csi.huawei.com	- Use the default value. - For the CCE Agile platform, modify this field. For example, csi.oceanstor.com.

Parameter	Description	Mandatory	Default Value	Remarks
<code>prometheusCollectorSSL.enabled</code>	Whether to enable the HTTPS service. After it is enabled, the Prometheus plug-in will provide the HTTPS service.	Yes	true	The certificate path must be specified after the service is enabled. For details about how to configure the certificate, see Configuring the HTTPS Service .
<code>prometheusCollectorSSL.certPath</code>	HTTPS certificate path after the Prometheus plug-in provides the HTTPS service.	This parameter is mandatory when <code>prometheusCollectorSSL.enabled</code> is set to true .	-	The path must be a relative path in the Helm working path.
<code>prometheusCollectorSSL.keyPath</code>	HTTPS key path after the Prometheus plug-in provides the HTTPS service.	This parameter is mandatory when <code>prometheusCollectorSSL.enabled</code> is set to true .	-	The path must be a relative path in the Helm working path.
<code>prometheusNodeSelector</code>	Node selector of csm-prometheus-service. After this parameter is set, csm-prometheus-service will be scheduled only to a node with the label.	No	-	For details about the node selector, see Assign Pods to Nodes .

Parameter	Description	Mandatory	Default Value	Remarks
prometheusCollector.tolerations	Taint tolerations of csm-prometheus-service. After this parameter is set, the scheduler allows csm-prometheus-service to be scheduled to a node with the corresponding taint.	No	For details, see the / helm/huawei-csm/values.yaml file in the software package.	For details about taints and tolerations, see Taints and Tolerations .
prometheusCollector.affinity	Node affinity of csm-prometheus-service. After this parameter is set, csm-prometheus-service will be preferentially scheduled to a node with the label.	No	For details, see the / helm/huawei-csm/values.yaml file in the software package.	For details about node affinity, see Assigning Pods to Nodes .
prometheusCollector.kubeAPIQps	Number of requests that can be sent by the client to kube-apiserver per second.	No	5	
prometheusCollector.kubeAPIBurst	Maximum number of requests that can be sent by the client in a short period.	No	10	
storageTopology.enabled	Whether to enable the storage topology display service.	Yes	true	Only OceanStor 6.1.7/OceanStor Dorado 6.1.7 and later versions are supported.
storageTopology.retryMaxDelay	Maximum retry delay of a topology resource synchronization task.	No	5m	You are advised to use the default value. The minimum retry delay is 5 seconds.

Parameter	Description	Mandatory	Default Value	Remarks
storageTopo.pvRetryMaxDelay	Maximum retry delay of PV resource creation and topo resource deletion tasks	No	1m	You are advised to use the default value. The minimum retry delay is 5 seconds.
storageTopo.podRetryMaxDelay	Maximum retry delay of the topo resource label update task for Pod resources.	No	1m	You are advised to use the default value. The minimum retry delay is 5 seconds.
storageTopo.resyncPeriod	Interval for refreshing topo resources.	No	15m	You are advised to use the default setting. The minimum refresh interval is 5 minutes.
storageTopo.nodeSelector	Node selector of csm-storage-service. After this parameter is set, csm-storage-service will be scheduled only to a node with the label.	No	-	For details about the node selector, see Assign Pods to Nodes .
storageTopo.tolerations	Taint tolerations of csm-storage-service. After this parameter is set, the scheduler allows csm-storage-service to be scheduled to a node with the corresponding taint.	No	For details, see the <code>/helm/huawei-csm/values.yaml</code> file in the software package.	For details about taints and tolerations, see Taints and Tolerations .

Parameter	Description	Mandatory	Default Value	Remarks
storageTopo.affinity	Node affinity of csm-storage-service. After this parameter is set, csm-storage-service will be preferentially scheduled to a node with the label.	No	For details, see the / helm/huawei-csm/values.yaml file in the software package.	For details about node affinity, see Assigning Pods to Nodes .
storageTopo.kubeAPIQps	Number of requests that can be sent by the client to kube-apiserver per second.	No	5	
storageTopo.kubeAPIBurst	Maximum number of requests that can be sent by the client in a short period.	No	10	

The **images** configuration items are used to configure the image information required by CSM.

Table 3-5 image configuration items

Parameter	Description	Mandatory	Default Value
prometheusCollector	Image of the Prometheus storage data collection plugin.	Yes	csm-prometheus-collector:2.7.0
topoService	Resource topology service image.	Yes	csm-topo-service:2.7.0
containerMonitorInterface	Container monitoring interface image.	Yes	csm-cmi:2.7.0
livenessProbe	Liveness probe service image.	Yes	csm-liveness-probe:2.7.0

The **cluster** configuration item mainly describes the Kubernetes cluster information.

Table 3-6 cluster configuration item

Parameter	Description	Mandatory	Default Value
cluster.name	Custom cluster name.	Yes	kubernetes

The **containerResourcesSet** configuration items are used to configure the container resources of each Pod.

Table 3-7 containerResourcesSet configuration items

Parameter	Description	Mandatory	Default Value
prometheusService.livenessProbe.requests.memory	Minimum memory resource of the livenessProbe container in a Prometheus Pod.	Yes	128Mi
prometheusService.livenessProbe.limits.cpu	Maximum CPU resource of the livenessProbe container in a Prometheus Pod.	Yes	100m
prometheusService.livenessProbe.limits.memory	Maximum memory resource of the livenessProbe container in a Prometheus Pod.	Yes	128Mi
prometheusService.prometheusCollector.requests.cpu	Minimum CPU resource of the prometheusCollector container in a Prometheus Pod.	Yes	50m
prometheusService.prometheusCollector.requests.memory	Minimum memory resource of the prometheusCollector container in a Prometheus Pod.	Yes	128Mi
prometheusService.prometheusCollector.limits.cpu	Maximum CPU resource of the prometheusCollector container in a Prometheus Pod.	Yes	300m

Parameter	Description	Mandatory	Default Value
prometheusService.prometheusCollector.limits.memory	Maximum memory resource of the prometheusCollector container in a Prometheus Pod.	Yes	512Mi
prometheusService.cmiController.requests.cpu	Minimum CPU resource of the cmiController container in a Prometheus Pod.	Yes	50m
prometheusService.cmiController.requests.memory	Minimum memory resource of the cmiController container in a Prometheus Pod.	Yes	128Mi
prometheusService.cmiController.limits.cpu	Maximum CPU resource of the cmiController container in a Prometheus Pod.	Yes	300m
prometheusService.cmiController.limits.memory	Maximum memory resource of the cmiController container in a Prometheus Pod.	Yes	512Mi
storageService.livenessProbe.requests.cpu	Minimum CPU resource of the livenessProbe container in a storage Pod.	Yes	10m
storageService.livenessProbe.requests.memory	Minimum memory resource of the livenessProbe container in a storage Pod.	Yes	128Mi
storageService.livenessProbe.limits.cpu	Maximum CPU resource of the livenessProbe container in a storage Pod.	Yes	100m
storageService.livenessProbe.limits.memory	Maximum memory resource of the livenessProbe container in a storage Pod.	Yes	128Mi
storageService.cmiController.requests.cpu	Minimum CPU resource of the cmiController container in a storage Pod.	Yes	50m

Parameter	Description	Mandatory	Default Value
storageService.cmiController.requests.memory	Minimum memory resource of the cmiController container in a storage Pod.	Yes	128Mi
storageService.cmiController.limits.cpu	Maximum CPU resource of the cmiController container in a storage Pod.	Yes	300m
storageService.cmiController.limits.memory	Maximum memory resource of the cmiController container in a storage Pod.	Yes	512Mi
storageService.topoService.requests.cpu	Minimum CPU resource of the topoService container in a storage Pod.	Yes	50m
storageService.topoService.requests.memory	Minimum memory resource of the topoService container in a storage Pod.	Yes	128Mi
storageService.topoService.limits.cpu	Maximum CPU resource of the topoService container in a storage Pod.	Yes	300m
storageService.topoService.limits.memory	Maximum memory resource of the topoService container in a storage Pod.	Yes	512Mi

Table 3-8 service configuration items

Parameter	Description	Mandatory	Default Value
service.ipFamilyPolicy	IP protocol stack that can be configured for the NodePort service of csm-prometheus-service. The following options can be configured: SingleStack , PreferDualStack , and RequireDualStack . For details about the parameter, visit https://kubernetes.io/docs/concepts/services-networking/dual-stack/#services .	No	SingleStack
service.ipFamilies	List of supported IP protocols. The value can be IPv4 or IPv6 .	No	IPv4

----End

4 Installation and Deployment

[4.1 Installation and Deployment Using Helm](#)

[4.2 Manual Installation and Deployment](#)

4.1 Installation and Deployment Using Helm

4.1.1 Installing the Software

Prerequisites

- Helm has been installed on the master node.
- The **values.yaml** file has been configured. For details, see [3.4 Preparing the Configuration File](#).

Preparations

For the OpenShift platform, run the following commands to create the **SecurityContextConstraints** resource.

1. Run the **vi huawei-csm-scc.yaml** command to create a **SecurityContextConstraints** file.

```
# vi huawei-csm-scc.yaml
allowHostDirVolumePlugin: true
allowHostIPC: false
allowHostNetwork: true
allowHostPID: false
allowHostPorts: false
allowPrivilegeEscalation: false
allowPrivilegedContainer: false

apiVersion: security.openshift.io/v1
kind: SecurityContextConstraints
metadata:
  name: huawei-csm-scc
runAsUser:
  type: RunAsAny
seLinuxContext:
  type: RunAsAny
fsGroup:
  type: RunAsAny
```

```
users:
- system:serviceaccount:huawei-csm:csm-prometheus-sa
- system:serviceaccount:huawei-csm:csm-storage-sa
volumes:
- hostpath
- emptyDir
- persistentVolumeClaim
- secret
- configMap
```

2. Run the **oc create -f huawei-csm-scc.yaml** command to create **SecurityContextConstraints**.

```
# oc create -f huawei-csm-scc.yaml
securitycontextconstraints.security.openshift.io/huawei-csm-scc created
```

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the **cd /opt/huawei-csm/helm/huawei-csm** command to go to the Helm working directory.
- Step 3** Run the **helm install huawei-csm ./ -n huawei-csm --create-namespace** command to install CSM services.

```
# helm install huawei-csm ./ -n huawei-csm --create-namespace
NAME: huawei-csm
LAST DEPLOYED: Tue Aug 8 23:11:18 2023
NAMESPACE: huawei-csm
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

- Step 4** Run the **kubectl get pod -n huawei-csm** command to check whether the services are started.

```
# kubectl get pod -n huawei-csm
NAME                                READY   STATUS    RESTARTS   AGE
csm-prometheus-service-86c795d68-b5xjg  2/2     Running   0          5s
csm-storage-service-85485fd75f-9wg8m    2/2     Running   0          4s
```

----End

4.1.2 Installing the Software on CCE Agile

4.1.2.1 Creating a Helm Installation Package

To install Huawei CSM on the CCE Agile platform, you need to create a Helm installation package. This section describes how to create a Helm installation package.

Prerequisites

- Helm has been installed on a node server.
- The **values.yaml** file required for installing CSM has been prepared. For details, see [3.4 Preparing the Configuration File](#).

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any node where Helm is deployed through the management IP address.
- Step 2** Run the `cd /opt/huawei-csm/helm/` command to go to the Helm working directory.
- Step 3** Run the `helm package huawei-csm/ -d ./` command to create a Helm installation package. This command will generate the installation package to the current path.
- ```
helm package huawei-csm/ -d ./
Successfully packaged chart and saved it to: huawei-csm-2.7.0.tgz
```
- End

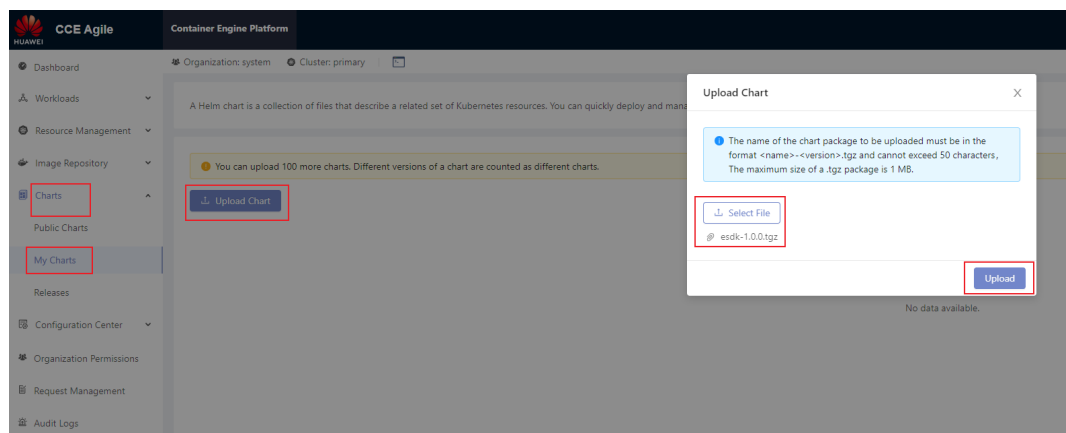
### 4.1.2.2 Installing Huawei CSM

#### Prerequisites

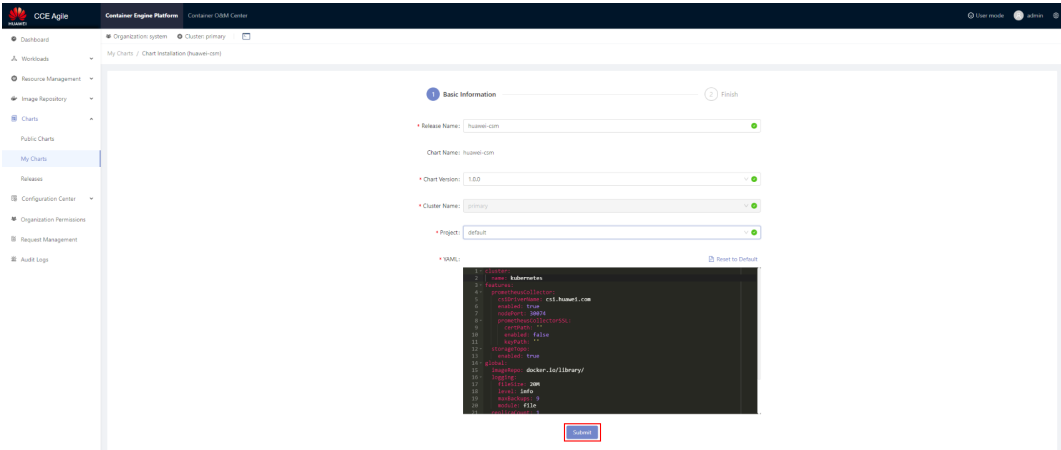
A Huawei CSM Helm installation package has been created. For details, see [4.1.2.1 Creating a Helm Installation Package](#).

#### Procedure

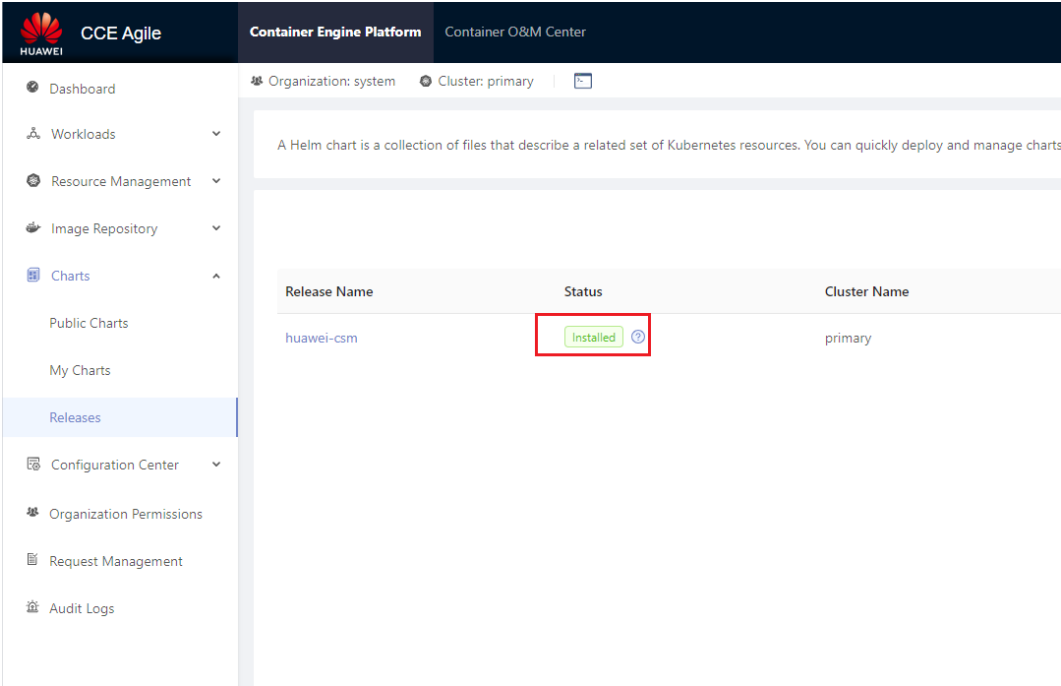
- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node where the CCE Agile platform is deployed through the management IP address.
- Step 2** Run the `kubectl create namespace huawei-csm` command to create a namespace for deploying Huawei CSM. *huawei-csm* indicates the custom namespace.
- ```
# kubectl create namespace huawei-csm
```
- Step 3** Export the Helm installation package. For details, see [4.1.2.1 Creating a Helm Installation Package](#).
- Step 4** On the home page, choose **Charts > My Charts > Upload Chart**. The **Upload Chart** dialog box is displayed. Import the exported Helm installation package to the CCE Agile platform.



- Step 5** After the installation package is uploaded, choose **Charts > My Charts**. On the **My Charts** page that is displayed, choose **Install > Submit**. The chart release name can be customized.



Step 6 On the home page, choose **Charts > Releases** and select the project specified during installation. After the installation is successful, **Installed** is displayed in the **Status** column.



----End

4.1.3 Upgrading the Software

Scenario

When upgrading the CSM service version, perform the operations described in this section.

Prerequisites

You have deployed CSM using Helm.

Precautions

During the upgrade, if the **values.yaml** and **update-value.yaml** files contain the same parameter settings, the parameters in the **update-value.yaml** file are preferentially used.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Import the new images to the corresponding node. For details, see [3.2 Uploading Images to the Image Repository](#).
- Step 3** (Optional) Run the **kubectl delete validatingWebhookConfiguration topo-service.xuanwu.huawei.io** command to delete the webhook resource. If the **topo-service.xuanwu.huawei.io** resource does not exist, skip this step.
- ```
kubectl delete validatingWebhookConfiguration topo-service.xuanwu.huawei.io
```
- Step 4** Run the **cd /opt/huawei-csm/helm/huawei-csm/** command to go to the Helm working directory.
- Step 5** Run the **kubectl apply -f crds/** command to update crd resources.
- ```
# kubectl apply -f crds/  
customresourcedefinition.apiextensions.k8s.io/resource-topologies.xuanwu.huawei.io configured
```
- Step 6** Run the **helm get values huawei-csm -n huawei-csm -a > update-value.yaml** command to obtain the original service configuration file.
- Step 7** Run the **vi update-value.yaml** command to open the file and update the parameter values as required. After the modification is complete, press **Esc** and enter **:wq!** to save the modification. For details, see [3.4 Preparing the Configuration File](#).
- Step 8** Run the **helm upgrade huawei-csm ./ -n huawei-csm -f ./values.yaml -f update-value.yaml --wait --timeout 2m** command to upgrade CSM services. If **Release "huawei-csm" has been upgraded** is displayed in the command output, the CSM services are successfully upgraded.
- ```
helm upgrade huawei-csm ./ -n huawei-csm -f ./values.yaml -f update-value.yaml --wait --timeout 2m
Release "huawei-csm" has been upgraded. Happy Helming!
NAME: huawei-csm
LAST DEPLOYED: Wed Aug 9 04:19:10 2023
NAMESPACE: huawei-csm
STATUS: deployed
REVISION: 3
TEST SUITE: None
```
- Step 9** (Optional) Run the **kubectl get rt | grep topo- | awk '{print "kubectl delete rt "\$1}' | sh** command to delete the residual topo resources of the source version. If there is no residual topo resource of the source version in the cluster, skip this step.
- End

## 4.1.4 Rolling Back the Software

### Prerequisites

You have deployed and upgraded CSM using Helm.

## Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the `cd /opt/huawei-csm/helm/huawei-csm/` command to go to the Helm working directory.
- Step 3** Run the `helm history huawei-csm -n huawei-csm` command to query the historical versions of the services deployed using Helm.

```
helm history huawei-csm -n huawei-csm
REVISION UPDATED STATUS CHART APP VERSION
DESCRIPTION
1 Tue Aug 8 23:11:18 2023 superseded huawei-csm-2.7.0 1.0.0 Install
complete
2 Wed Aug 9 04:19:10 2023 deployed huawei-csm-2.7.0 1.0.0 Upgrade complete
```

- Step 4** Run the `helm rollback huawei-csm revision-number -n huawei-csm --wait --timeout 2m` command to roll back the CSM services to the specified version. If **Rollback was a success** is displayed in the command output, the CSM services are successfully rolled back to the specified version.

In the preceding command, *revision-number* indicates a version number queried in [Step 3](#). For example, the version is **1**.

```
helm rollback huawei-csm 1 -n huawei-csm --wait --timeout 2m
Rollback was a success! Happy Helming!
```

----End

## 4.1.5 Uninstalling the Software

### Prerequisites

- You have deployed CSM using Helm.
- You have deleted resources created by CSM.

## Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the `cd /opt/huawei-csm/helm/huawei-csm/` command to go to the Helm working directory.
- Step 3** Run the `helm uninstall huawei-csm -n huawei-csm` command to uninstall CSM services. If **release "huawei-csm" uninstalled** is displayed in the command output, the services are successfully uninstalled.

```
helm uninstall huawei-csm -n huawei-csm
release "huawei-csm" uninstalled
```

- Step 4** (Optional) Run the `kubectl delete validatingWebhookConfiguration topo-service.xuanwu.huawei.io` command to delete the webhook resource. If the **topo-service.xuanwu.huawei.io** resource does not exist, skip this step.

```
kubectl delete validatingWebhookConfiguration topo-service.xuanwu.huawei.io
```

- Step 5** (Optional) Run the `kubectl delete -f crds/` command to clear crd resources.

**NOTICE**

- If Huawei CSM is no longer used and related resource objects of Huawei CSM have been cleared from the environment, perform this step. Otherwise, skip this step.
- Before deleting crd resources, ensure that the resources created using Huawei CSM have been cleared. For details about the crd resource types, see the files in the `/opt/huawei-csm/helm/huawei-csm/crds` directory.
- Deleting a crd resource will clear all resources associated with the crd. Exercise caution when performing this operation.
- If a message similar to **<Error from server (NotFound): error when deleting "crds/xuanwu.huawei.io\_resourcetopologies.yaml": customresourcedefinitions.apiextensions.k8s.io "resourcetopologies.xuanwu.huawei.io" not found>** is displayed, the crd resource has been uninstalled. In this case, ignore the message.

```
kubectl delete -f crds/
customresourcedefinition.apiextensions.k8s.io "resourcetopologies.xuanwu.huawei.io" deleted
```

**Step 6** Run the **kubectl delete ns huawei-csm** command to delete the namespace.

**NOTICE**

Deleting a namespace will clear all resources in the namespace. Exercise caution when performing this operation.

**Step 7** If the OpenShift platform is used, run the **oc delete securitycontextconstraints huawei-csm-scc** command to delete the **SecurityContextConstraints** resource. Otherwise, skip this step.

```
oc delete securitycontextconstraints huawei-csm-scc
securitycontextconstraints.security.openshift.io "huawei-csm-scc" deleted
```

----End

## 4.1.6 Uninstalling the Software from CCE Agile

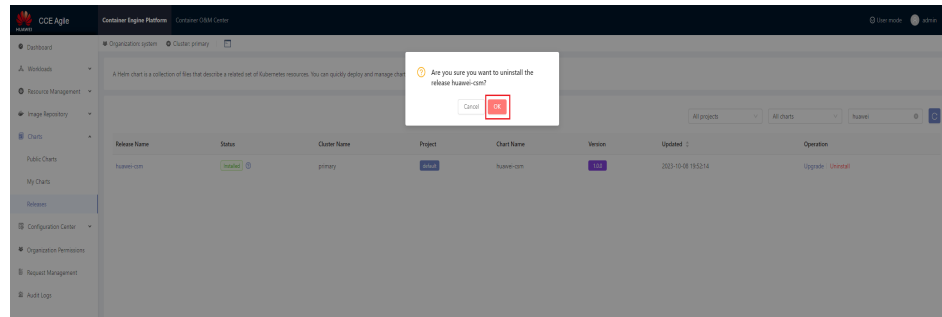
This section describes how to uninstall Huawei CSM from the CCE Agile platform. The following uses CCE Agile v22.3.2 as an example.

### Procedure

**Step 1** Log in to the CCE Agile platform.

**Step 2** On the home page, choose **Charts > Releases**. The **Releases** page is displayed.

**Step 3** Select a Huawei CSM release and click **Uninstall**. In the displayed dialog box, click **OK**.



----End

## 4.2 Manual Installation and Deployment

### 4.2.1 Manually Installing the Software

#### Preparations

For the OpenShift platform, run the following commands to create the **SecurityContextConstraints** resource.

1. Run the **vi huawei-csm-scc.yaml** command to create a **SecurityContextConstraints** file.

```
vi huawei-csm-scc.yaml
allowHostDirVolumePlugin: true
allowHostIPC: false
allowHostNetwork: true
allowHostPID: false
allowHostPorts: false
allowPrivilegeEscalation: false
allowPrivilegedContainer: false

apiVersion: security.openshift.io/v1
kind: SecurityContextConstraints
metadata:
 name: huawei-csm-scc
runAsUser:
 type: RunAsAny
seLinuxContext:
 type: RunAsAny
fsGroup:
 type: RunAsAny
users:
- system:serviceaccount:huawei-csm:csm-prometheus-sa
- system:serviceaccount:huawei-csm:csm-storage-sa
volumes:
- hostpath
- emptyDir
- persistentVolumeClaim
- secret
- configMap
```

2. Run the **oc create -f huawei-csm-scc.yaml** command to create **SecurityContextConstraints**.

```
oc create -f huawei-csm-scc.yaml
securitycontextconstraints.security.openshift.io/huawei-csm-scc created
```

## Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the **kubectl create namespace huawei-csm** command to create a namespace named **huawei-csm**.

```
kubectl create namespace huawei-csm
namespace/huawei-csm created
```

**Step 3** Run the **cd /opt/huawei-csm/manual/huawei-csm** command to go to the Huawei CSM installation directory.

**Step 4** Run the **kubectl apply -f crds/** command to install crd resources.

```
kubectl apply -f crds/
customresourcedefinition.apiextensions.k8s.io/resourcepolicies.xuanwu.huawei.io created
```

**Step 5** Run the **cd /opt/huawei-csm/manual/huawei-csm/templates** command to go to the Huawei CSM working directory.

**Step 6** (Optional) If the single-copy deployment mode is used, skip this step. If the multi-copy deployment mode is used, you need to manually change the number of Deployment resource copies in all files in the **templates** directory and set all **enable-leader-election** parameters to **true**.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 app: csm-storage-service
 name: csm-storage-service
 namespace: huawei-csm
spec:
 progressDeadlineSeconds: 600
 replicas: 2
 ...
 args:
 - --enable-leader-election=true
 ...
```

**Step 7** (Optional) To configure the node selector of the CSM service, manually uncomment the **nodeSelector** field of the Deployment resources in all files in the **templates** directory and configure the corresponding node matching tag.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: csm-prometheus-service
 ...
spec:
uncomment if you wish to configure selection constraints for csm-prometheus-service pods
 nodeSelector:
 kubernetes.io/hostname: <host-name>
 ...
```

**Step 8** (Optional) To enable the Prometheus plug-in to provide the HTTPS service, manually configure the **prometheus-ssl.yaml** certificate file by following the instructions in [Table 4-1](#). Run the **kubectl apply -f prometheus-ssl.yaml** command to create a certificate file. Manually uncomment the certificate-related fields in the **csm-prometheus.yaml** file and change the value of **use-https** to **true**. If the default HTTP service is used, skip this step.

```
kubectl apply -f prometheus-ssl.yaml
secret/prometheus-ssl created
```

**Table 4-1** prometheus-ssl parameters

| Parameter    | Description                                   | Mandatory | Default Value |
|--------------|-----------------------------------------------|-----------|---------------|
| data.tls.crt | Certificate information encoded using Base64. | Yes       | -             |
| data.tls.key | Private key information encoded using Base64. | Yes       | -             |

```
prometheus-ssl.yaml
...
 containers:
 - name: prometheus-collector
...
 args:
 - --use-https=true # modify the value to "true" if configured the SSL cert
...
 volumeMounts:
uncomment if configured the SSL cert
 - name: secret-volume
 mountPath: /etc/secret-volume
 readOnly: true
 livenessProbe:
 failureThreshold: 5
 httpGet:
uncomment if configured the SSL cert
 scheme: HTTPS
 path: /healthz
 port: 8887
...
 volumes:
uncomment if configured the SSL cert
 - name: secret-volume
 secret:
 secretName: prometheus-ssl
 defaultMode: 0400
```

**Step 9** Run the **kubectl apply -f csm-prometheus.yaml** command to deploy the csm-prometheus service.

```
kubectl apply -f csm-prometheus.yaml
serviceaccount/csm-prometheus-sa created
clusterrole.rbac.authorization.k8s.io/prometheus-collector-role created
clusterrolebinding.rbac.authorization.k8s.io/prometheus-collector-binding created
clusterrole.rbac.authorization.k8s.io/cmi-collector-role created
clusterrolebinding.rbac.authorization.k8s.io/cmi-collector-binding created
deployment.apps/csm-prometheus-service created
service/csm-prometheus-service created
```

**Step 10** Run the **kubectl apply -f csm-storage.yaml** command to deploy the csm-storage service.

```
kubectl apply -f csm-storage.yaml
serviceaccount/csm-storage-sa created
clusterrole.rbac.authorization.k8s.io/topo-service-role created
clusterrole.rbac.authorization.k8s.io/cmi-controller-role created
clusterrolebinding.rbac.authorization.k8s.io/topo-service-binding created
clusterrolebinding.rbac.authorization.k8s.io/cmi-controller-binding created
deployment.apps/csm-storage-service created
service/csm-storage-service created
```

**Step 11** Run the **kubectl get pod -n huawei-csm** command to check the Pod creation result. If the following information is displayed, the Huawei CSM services are successfully created.

```
kubectl get pod -n huawei-csm
NAME READY STATUS RESTARTS AGE
csm-prometheus-service-7c8bd8fd89-lbfvn 3/3 Running 0 3m21s
csm-storage-service-747dbc5cbd-n24j8 3/3 Running 0 2m19s
```

----End

## 4.2.2 Manually Updating the Software

Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 1** Manually uninstall Huawei CSM of the earlier version by following the instructions in [4.2.3 Manually Uninstalling the Software](#).

**Step 2** Go to the `/opt/huawei-csm/manual/huawei-csm` directory where the new installation package is stored and run the **kubectl apply -f crds/** command to update crd resources.

```
kubectl apply -f crds/
customresourcedefinition.apiextensions.k8s.io/resource-topologies.xuanwu.huawei.io configured
```

**Step 3** Install Huawei CSM of the current version. For details, see [4.2.1 Manually Installing the Software](#).

----End

## 4.2.3 Manually Uninstalling the Software

### Prerequisites

Huawei CSM has been installed and is running properly.

### Uninstalling the csm-prometheus Service

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the **cd /opt/huawei-csm/manual/huawei-csm/templates** command to go to the Huawei CSM working directory.

**Step 3** Run the **kubectl delete -f csm-prometheus.yaml** command to uninstall the csm-prometheus service.

```
kubectl delete -f csm-prometheus.yaml
serviceaccount "csm-prometheus-sa" deleted
clusterrole.rbac.authorization.k8s.io "prometheus-collector-role" deleted
clusterrolebinding.rbac.authorization.k8s.io "prometheus-collector-binding" deleted
clusterrole.rbac.authorization.k8s.io "cmi-collector-role" deleted
clusterrolebinding.rbac.authorization.k8s.io "cmi-collector-binding" deleted
deployment.apps "csm-prometheus-service" deleted
service "csm-prometheus-service" deleted
```

**Step 4** (Optional) If no certificate file is configured, skip this step. Run the **kubectl delete -f prometheus-ssl.yaml** command to delete the certificate secret file.

```
kubectl delete -f prometheus-ssl.yaml
secret "prometheus-ssl" deleted
```

----End



## Uninstalling the csm-storage Service

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the **cd /opt/huawei-csm/manual/huawei-csm/templates** command to go to the Huawei CSM working directory.
- Step 3** Run the **kubectl delete -f csm-storage.yaml** command to uninstall the csm-storage service.

```
kubectl delete -f csm-storage.yaml
serviceaccount "csm-storage-sa" deleted
clusterrole.rbac.authorization.k8s.io "topo-service-role" deleted
clusterrole.rbac.authorization.k8s.io "cmi-controller-role" deleted
clusterrolebinding.rbac.authorization.k8s.io "topo-service-binding" deleted
clusterrolebinding.rbac.authorization.k8s.io "cmi-controller-binding" deleted
deployment.apps "csm-storage-service" deleted
service "csm-storage-service" deleted
```

----End

## Uninstalling Other Resources

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the **cd /opt/huawei-csm/manual/huawei-csm** command to go to the Huawei CSM installation directory.
- Step 3** Run the **kubectl delete configmap -n huawei-csm huawei-csm-version** command to delete the **configmap** resources.
- Step 4** (Optional) If Huawei CSM is deployed in single-copy mode, skip this step. Run the **kubectl delete lease -n huawei-csm resource-topology** command to delete the **lease** resources.
- Step 5** (Optional) Run the **kubectl delete -f crds/** command to clear crd resources.

### NOTICE

- If Huawei CSM is no longer used and related resource objects of Huawei CSM have been cleared from the environment, perform this step. Otherwise, skip this step.
- Before deleting crd resources, ensure that the resources created using Huawei CSM have been cleared. For details about the crd resource types, see the files in the **/opt/huawei-csm/manual/huawei-csm/crds** directory.
- Deleting a crd resource will clear all resources associated with the crd. Exercise caution when performing this operation.
- If a message similar to **<Error from server (NotFound): error when deleting "crds/xuanwu.huawei.io\_resourcetopologies.yaml": customresourcedefinitions.apiextensions.k8s.io "resourcetopologies.xuanwu.huawei.io" not found>** is displayed, the crd resource has been uninstalled. In this case, ignore the message.

```
kubectl delete -f crds/
customresourcedefinition.apiextensions.k8s.io "resourcetopologies.xuanwu.huawei.io" deleted
```

**Step 6** Run the **kubectl delete ns huawei-csm** command to delete the namespace.

---

#### NOTICE

Deleting a namespace will clear all resources in the namespace. Exercise caution when performing this operation.

---

**Step 7** If the OpenShift platform is used, run the **oc delete securitycontextconstraints huawei-csm-scc** command to delete the **SecurityContextConstraints** resource. Otherwise, skip this step.

```
oc delete securitycontextconstraints huawei-csm-scc
securitycontextconstraints.security.openshift.io "huawei-csm-scc" deleted
```

----End

# 5 Configuring Prometheus

---

[5.1 Installing Prometheus](#)

[5.2 Configuring the Prometheus Service](#)

[5.3 Configuring the Prometheus Dashboard](#)

## 5.1 Installing Prometheus

Prometheus is an open-source systems monitoring and alerting toolkit. Prometheus joined the Cloud Native Computing Foundation (CNCF) in 2016 as the second hosted project, after Kubernetes.

For details about how to obtain and install Prometheus, [click here](#).

For other information about Prometheus, [click here](#).

## 5.2 Configuring the Prometheus Service

### Configuration Description

Prometheus invokes the interfaces provided by the plug-in to monitor storage data. No matter Prometheus is deployed in Kubernetes or on an independent server, a configuration file is required to start Prometheus and monitor the plug-in.

This section only describes how to add configurations related to the plug-in to the Prometheus configuration file. For details about other configurations, [click here](#) to see the Prometheus official document.

**Step 1** Modify the Prometheus configuration file. Add the storage backend monitoring items created using CSM to the **scrape\_configs** section.

#### NOTE

This configuration file is the configuration file of the Prometheus platform. Its location varies depending on the deployment mode. For details, [click here](#) to see the Prometheus official document.

**Step 2** Add resource monitoring items. The configuration template is as follows:

```
scrape_configs:
- job_name: *****
 scrape_interval: ***
 scrape_timeout: ***
 metrics_path: /object/**
 scheme: http/https
 params:
 controller: [""]
 storagepool: [""]
 filesystem: [""]
 lun: [""]
 array: [""]
 pv: [""]
 vstore: [""]
 tls_config:
 ca_file: *****
 cert_file: *****
 key_file: *****
 static_configs:
 - targets: ['*.*.*:<Port number>']
```

**Table 5-1** describes the parameters in the **scrape\_configs** section.

**Table 5-1** Parameter description

| Parameter       | Description                                                                                                                                                                                                                                                                                                                                                               | Example                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| job_name        | Monitoring job name.                                                                                                                                                                                                                                                                                                                                                      | job_name: OceanStor-Monitor    |
| scheme          | Web request mode. If this parameter is not set, the default value <b>http</b> is used. After <b>tls_config</b> is configured in the plug-in, set this parameter to <b>https</b> .                                                                                                                                                                                         | scheme: https                  |
| scrape_interval | Data scraping interval.                                                                                                                                                                                                                                                                                                                                                   | scrape_interval: 15m           |
| scrape_timeout  | Data scraping timeout. You are advised to set this parameter to be the same as the value of <b>scrape_interval</b> .                                                                                                                                                                                                                                                      | scrape_timeout: 15m            |
| metrics_path    | Data scraping path, in the format of <b>/object/**</b> . <b>***</b> indicates the storage backend name in the plug-in configuration file.<br><b>NOTE</b><br>Only users with the corresponding roles in the storage system group can query the metric data of the backend.<br><br>Such roles include the super administrator, administrator, and monitoring administrator. | metrics_path: /object/backend1 |

| Parameter  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Example                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| params     | <p>Monitored object of centralized storage. Currently, the following types are supported:</p> <ul style="list-style-type: none"><li>• <b>controller</b>: controller</li><li>• <b>storagepool</b>: storage pool</li><li>• <b>filesystem</b>: file system</li><li>• <b>lun</b>: LUN</li><li>• <b>array</b>: array</li><li>• <b>pv</b>: PV data of Kubernetes</li><li>• <b>vstore</b>: vStore quota information on the storage system</li></ul> <p><b>NOTE</b></p> <ul style="list-style-type: none"><li>• If you do not need a certain object, you can delete it.</li><li>• The value of an object parameter must be in the format of <code>[""]</code>.</li><li>• For details about object metric data, see <a href="#">Table 2-4</a>.</li></ul> | <p>controller: <code>[""]</code><br/>storagepool: <code>[""]</code><br/>filesystem: <code>[""]</code><br/>lun: <code>[""]</code><br/>array: <code>[""]</code><br/>pv: <code>[""]</code><br/>vstore: <code>[""]</code></p> |
| tls_config | <p>TLS configuration when <b>scheme</b> is set to <b>https</b>. This parameter is mandatory when <b>scheme</b> is set to <b>https</b>.</p> <ul style="list-style-type: none"><li>• <b>ca_file</b>: path of the CA certificate used to verify the API server certificate.</li><li>• <b>cert_file</b>: path of the certificate file used for client certificate authentication on the server.</li><li>• <b>key_file</b>: path of the key file used for client certificate authentication on the server.</li></ul>                                                                                                                                                                                                                                 | <p>ca_file: <code>/opt/huawei/ca.crt</code><br/>cert_file: <code>/opt/huawei/client.crt</code><br/>key_file: <code>/opt/huawei/client.key</code></p> <p><b>NOTE</b><br/>The paths can be customized.</p>                  |
| targets    | <p>Listening address (for example, the IP address of a host in the cluster deployed using CSM) and port exposed by the CSM plug-in.</p> <p>For Kubernetes deployment, the default port is 30074. The specific port is the one specified during plug-in deployment.</p> <p>For non-Kubernetes deployment, no default port is available. The specific port is the one specified during plug-in deployment.</p>                                                                                                                                                                                                                                                                                                                                    | <p><code>['192.168.1.1:30074']</code></p>                                                                                                                                                                                 |

 NOTE

For details about **scrape\_configs** parameters, [click here](#) to visit the Prometheus official website.

**Step 3** Add performance monitoring items. The configuration template is as follows:

```
scrape_configs:
- job_name: *****
 scrape_interval: ***
 scrape_timeout: ***
 metrics_path: /performance/**
 scheme: http/https
 params:
 controller: ['21,22,370']
 storagepool: ['21,22,370']
 lun: ['21,22,370']
 filesystem: ['182,524,525']
 pv: ['filesystem,lun']
 tls_config:
 ca_file: *****
 cert_file: *****
 key_file: *****
 static_configs:
 - targets: ['*.*.*:<Port number>']
```

**Table 5-2** describes the parameters in the **scrape\_configs** section.

**Table 5-2** Parameter description

| Parameter       | Description                                                                                                                                                                       | Example                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| job_name        | Monitoring job name.                                                                                                                                                              | job_name: OceanStor-Monitor |
| scheme          | Web request mode. If this parameter is not set, the default value <b>http</b> is used. After <b>tls_config</b> is configured in the plug-in, set this parameter to <b>https</b> . | scheme: https               |
| scrape_interval | Data scraping interval.                                                                                                                                                           | scrape_interval: 15m        |
| scrape_timeout  | Data scraping timeout. You are advised to set this parameter to be the same as the value of <b>scrape_interval</b> .                                                              | scrape_timeout: 15m         |

| Parameter    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Example                                                                                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metrics_path | <p>Data scraping path, in the format of <b>/performance/***</b>. *** indicates the storage backend name in the plug-in configuration file.</p> <p><b>NOTE</b></p> <p>Only users with the corresponding roles in the storage system group can query the metric data of the backend.</p> <p>Such roles include the super administrator, administrator, and monitoring administrator.</p>                                                                                                                                                                                                                                                                                                                                  | metrics_path: /performance/backend1                                                                                                                            |
| params       | <p>Currently, the following types are supported:</p> <ul style="list-style-type: none"><li>• <b>controller</b>: controller</li><li>• <b>storagepool</b>: storage pool</li><li>• <b>filesystem</b>: file system</li><li>• <b>lun</b>: LUN</li><li>• <b>pv</b>: PV data of Kubernetes</li></ul> <p><b>NOTE</b></p> <ul style="list-style-type: none"><li>• If you do not need a certain object, you can delete it.</li><li>• The object parameter value is used to specify the performance metric of the object. The format is 'Metric 1,Metric 2...'. If you do not need a certain metric, you can delete it.</li><li>• For details about performance metric data and mappings, see <a href="#">Table 2-5</a>.</li></ul> | controller: ['21,22,23,25,26,28,370']<br>storagepool: ['21,22,370']<br>lun: ['21,22,370']<br>filesystem: ['182,524,525']<br>pv: ['filesystem,lun']             |
| tls_config   | <p>TLS configuration when <b>scheme</b> is set to <b>https</b>. This parameter is mandatory when <b>scheme</b> is set to <b>https</b>.</p> <ul style="list-style-type: none"><li>• <b>ca_file</b>: path of the CA certificate used to verify the API server certificate.</li><li>• <b>cert_file</b>: path of the certificate file used for client certificate authentication on the server.</li><li>• <b>key_file</b>: path of the key file used for client certificate authentication on the server.</li></ul>                                                                                                                                                                                                         | ca_file: /opt/huawei/ca.crt<br>cert_file: /opt/huawei/client.crt<br>key_file: /opt/huawei/client.key<br><p><b>NOTE</b></p> <p>The paths can be customized.</p> |

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                   | Example               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| targets   | Listening address (for example, the IP address of a host in the cluster deployed using CSM) and port exposed by the CSM plug-in.<br>For Kubernetes deployment, the default port is 30074. The specific port is the one specified during plug-in deployment.<br>For non-Kubernetes deployment, no default port is available. The specific port is the one specified during plug-in deployment. | ['192.168.1.1:30074'] |

 **NOTE**

For details about **scrape\_configs** parameters, [click here](#) to visit the Prometheus official website.

**Step 4** To monitor multiple storage backends, repeat [Step 2](#) and [Step 3](#).

**Step 5** Restart the Prometheus service.

 **NOTE**

Prometheus is an open-source component. The startup mode varies depending on the deployment mode. For details, [click here](#) to see the Prometheus official document.

----End

## 5.3 Configuring the Prometheus Dashboard

**Step 1** In the address box of a browser, enter the IP address and port of the Prometheus service to log in to the Prometheus monitoring UI. The default port is **9090**.

**Step 2** Click the **Graph** tab, enter **huawei** in the search box, and select the desired monitoring item.

----End



# 6 Configuring Grafana

---

## 6.1 Installing Grafana

### 6.2 Using Grafana

## 6.1 Installing Grafana

Grafana is an open-source visualization platform that provides full support for Prometheus.

#### NOTE

- Grafana 2.5.0 (released on October 28, 2015) and later versions allow Prometheus to function as a Grafana data source. Currently, Huawei storage can interconnect with only Grafana 11.6.1 to 12.1.1.
- CCE Agile provides the Grafana component and has the default Prometheus data source. You can go to [Using Grafana to Monitor Storage](#) to perform the next step.

For details about how to install Grafana, see [the Grafana official documentation](#).

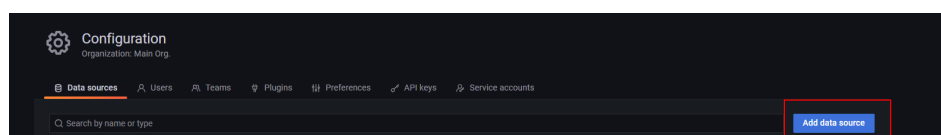
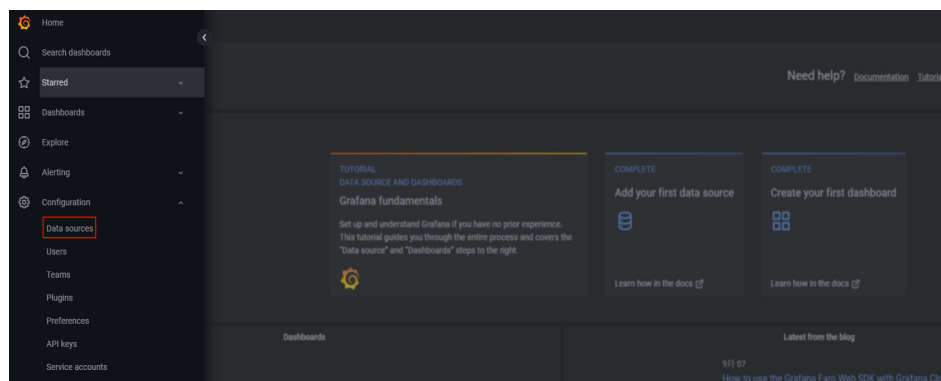
## 6.2 Using Grafana

#### NOTE

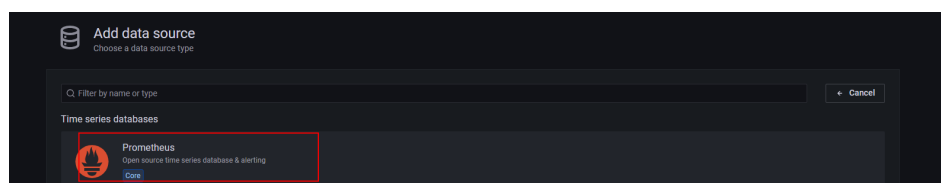
By default, Grafana listens on port 3000.

### Adding Prometheus and Configuring a Data Source

- Step 1** In the navigation pane on the left, choose **Configuration > Data sources** and click **Add data source** to add a data source.

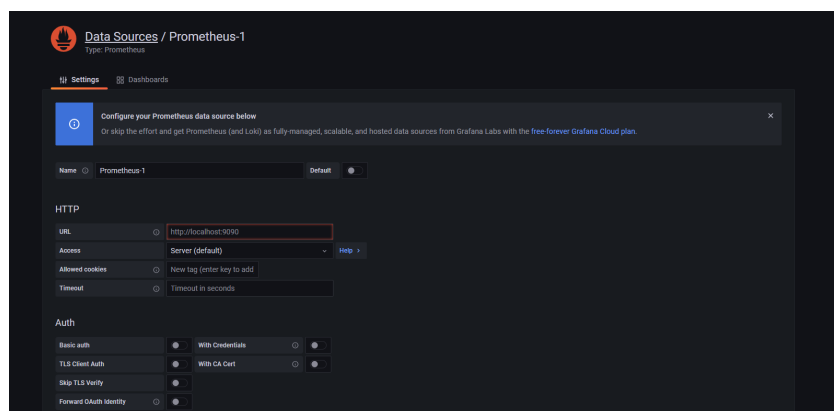


**Step 2** Select Prometheus and add a data source.



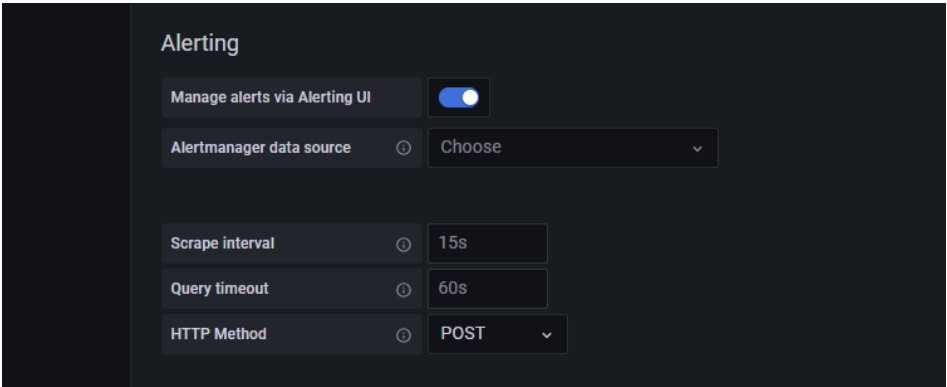
**Step 3** Configure data source information.

- **Name:** name of the data source.
- **URL:** IP address of the Prometheus server.
- **Access:** Set this parameter based on site requirements.
  - If this parameter is set to **Server**, a browser sends a request to the Grafana server and then the Grafana server sends a request to the Prometheus server to obtain data.
  - If this parameter is set to **Browser**, a browser directly sends a request to the Prometheus server to obtain data. (The cross-domain problem needs to be solved.)

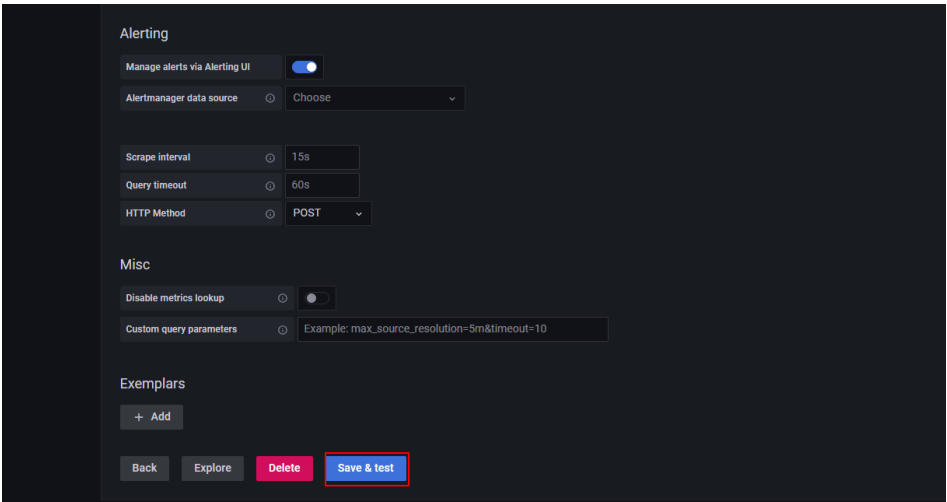


**Step 4** Set the HTTP method.

- **POST:** recommended when a large amount of data is queried.



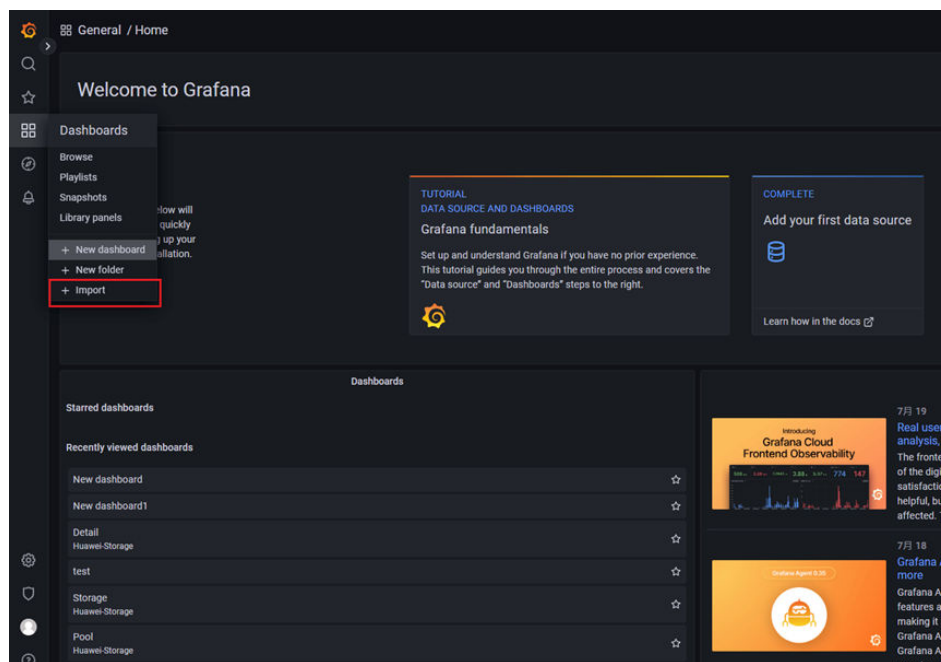
**Step 5** Click **Save & Test** to save the new data source.



----End

## Using Grafana to Monitor Storage

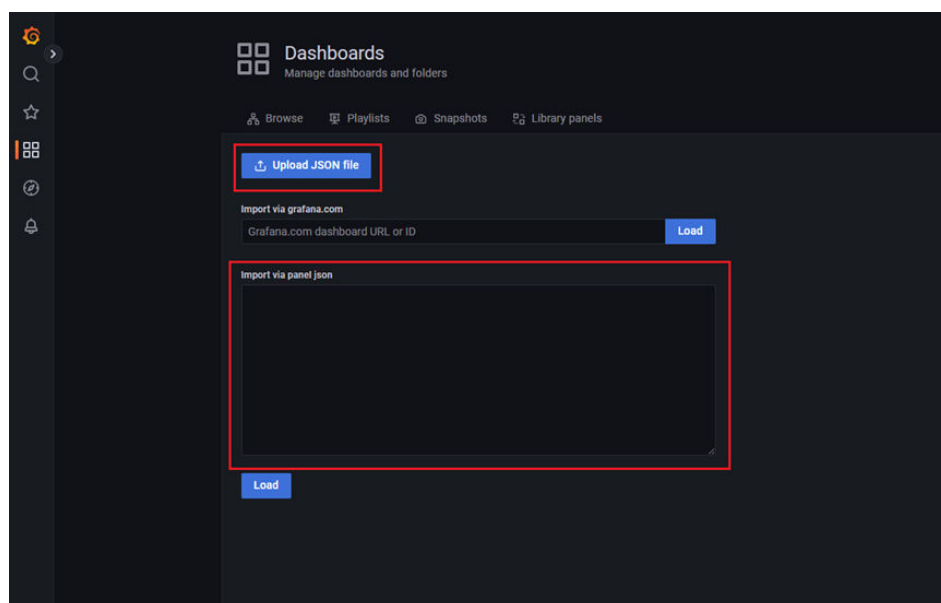
**Step 1** In the navigation pane on the left, choose **Dashboards > Import**.



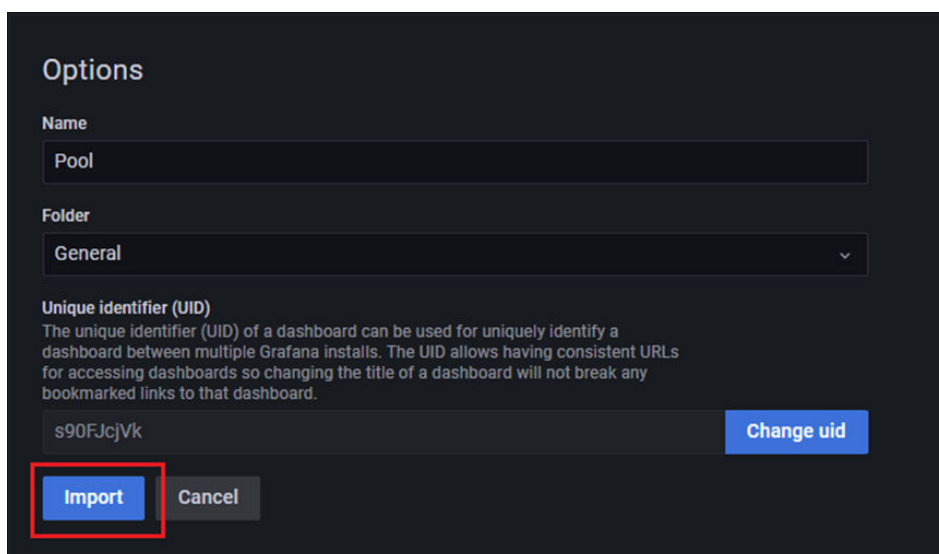
**Step 2** Click **Upload JSON file** to upload a JSON file or directly paste JSON data to load the existing dashboard. After the loading is successful, you can view the JSON file in the **helm/huawei-csm/grafana/OceanStor.json** directory in the path where the CSM software package is decompressed on the **Browse** tab page. The JSON file is a sample template.

#### NOTE

1. The **OceanStor.json** sample template aims to help users quickly understand the Grafana display effect. In the actual production environment, you are advised to customize Grafana dashboards based on service requirements, monitoring scope, and organization visualization specifications.
2. By default, this template displays all monitoring metrics supported by the CSM. Some panels depend on each other. Before importing the template, ensure that all metrics have been successfully collected so that the dashboard can be properly rendered and all information can be displayed.

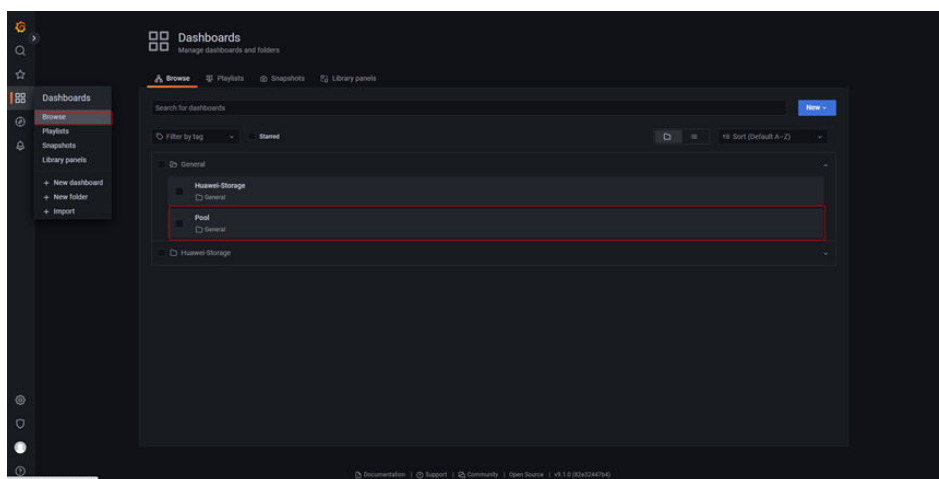


**Step 3** After setting the dashboard name and folder, click **Import**.



The image shows the 'Options' dialog for importing a dashboard in Grafana. The 'Name' field is set to 'Pool'. The 'Folder' is set to 'General'. Below these fields is a text box for the 'Unique identifier (UID)' containing 's90FJcjVk'. A blue 'Change uid' button is to the right of the UID field. At the bottom, there are two buttons: 'Import' (highlighted with a red rectangle) and 'Cancel'.

**Step 4** In the navigation pane on the left, choose **Dashboards > Browse**. On the **Browse** tab page, find the imported dashboard and click it.



**Step 5** In the **DataSource** drop-down list on the dashboard, select the configured data source.



 **NOTE**

The Huawei logo in the upper left corner of the template can be loaded only in a public network environment.

**----End**

# 7 Advanced Features

---

## 7.1 Configuring IPv6

## 7.1 Configuring IPv6

### 7.1.1 Configuring IPv6 During the Initial Installation of Huawei CSM

This section describes how to configure IPv6 during the initial installation of Huawei CSM.

#### Prerequisites

- The host environment and Kubernetes cluster environment support IPv6. For details, see [IPv4/IPv6 dual-stack of Kubernetes](#).
- Huawei CSI has been installed and services are running properly.

#### Procedure

- Step 1** Run the **vi values.yaml** command to open the configuration file and modify the **service.ipFamilyPolicy** and **service.ipFamilies** configuration items. For details about the parameters, see [Table 3-8](#).

The following is an example:

```
service:
 ipFamilyPolicy: SingleStack
 ipFamilies:
 - IPv6
```

- Step 2** Install and deploy Huawei CSM by following the instructions in [4 Installation and Deployment](#).

- Step 3** After Huawei CSM is installed, you can run the following command to check the service status in the **huawei-csm** namespace.

```
kubectl get service -n huawei-csm
```

The following is an example of the command output.

| NAME                   | TYPE     | CLUSTER-IP       | EXTERNAL-IP | PORT(S)        | AGE  |
|------------------------|----------|------------------|-------------|----------------|------|
| csm-prometheus-service | NodePort | fd00:10:96::7800 | <none>      | 8887:30074/TCP | 102d |

**Step 4** Run the following command to check the value of **ipFamilies** for the csm-prometheus-service service.

```
kubectl get svc -n huawei-csm csm-prometheus-service -o=jsonpath='{.spec.ipFamilies}'
```

The following is an example of the command output.

```
["IPv6"]
```

**Step 5** Run the following command to check the value of **ipFamilyPolicy** for the csm-prometheus-service service.

```
kubectl get svc -n huawei-csm csm-prometheus-service -o=jsonpath='{.spec.ipFamilyPolicy}'
```

The following is an example of the command output.

```
SingleStack
```

----End

## 7.1.2 Switching from IPv4 to IPv6 for Huawei CSM

This section describes how to switch from IPv4 to IPv6 when updating Huawei CSM.

### Prerequisites

- The host environment and Kubernetes cluster environment support IPv6. For details, see [IPv4/IPv6 dual-stack of Kubernetes](#).
- Huawei CSI has been installed and services are running properly.
- Huawei CSM has been installed and services are running properly.

### Procedure

**Step 1** Run the following command to obtain the original service configuration file.

```
helm get values huawei-csm -n huawei-csm -a > update-value.yaml
```

**Step 2** Run the **vi update-values.yaml** command to open the file obtained in [Step 1](#) and modify the **service.ipFamilyPolicy** and **service.ipFamilies** configuration items. For details about the parameters, see [Table 3-8](#).

The following is an example:

```
service:
 ipFamilyPolicy: SingleStack
 ipFamilies:
 - IPv6
```

**Step 3** Before switching to IPv6, run the following command to delete the service in the **huawei-csm** namespace.

```
kubectl delete service -n huawei-csm --all
```

**Step 4** Run the following command to update Huawei CSM. **update-values.yaml** is the file obtained in [Step 1](#).

```
helm upgrade huawei-csm ./ -n huawei-csm -f ./values.yaml -f update-value.yaml --wait --timeout 2m
```

**Step 5** Run the following command to query the updated parameters of the service.

```
kubectl get svc -n huawei-csm csm-prometheus-service -o=jsonpath='{.spec.ipFamilies}'
```



The expected output is as follows:

```
["IPv6"]
```

----**End**

# 8 Common O&M Guide

## 8.1 Collecting Logs

### 8.2 Viewing Version Information

## 8.1 Collecting Logs

### NOTICE

If you use the `oceanctl` tool to collect logs, the CPU usage on the master node may increase sharply in a short period of time. Therefore, determine whether to use this function based on the workload and CPU usage on the master node. You can also manually collect CSM logs in the log archive directory (`/var/log/huawei-csm/`) on each node.

### 8.1.1 Performing Check Before Collection

**Step 1** Use a remote access tool, such as PuTTY, to log in to the node where the `oceanctl` tool is installed in the Kubernetes cluster through the management IP address.

**Step 2** Run the `oceanctl version` command. The displayed version is **v4.12.0**.

```
$ oceanctl version
Oceanctl Version: v4.12.0
```

**Step 3** Run the `oceanctl --help` command. The following information is displayed.

```
$ oceanctl --help
A CLI tool for Ocean Storage in Kubernetes

Usage:
 oceanctl [command]

Available Commands:
 collect collect messages in Kubernetes
 create Create a resource to Ocean Storage in Kubernetes
 delete Delete one or more resources from Ocean Storage in Kubernetes
 get Get one or more resources from Ocean Storage in Kubernetes
 help Help about any command
 update Update a resource for Ocean Storage in Kubernetes
 version Print the version of oceanctl
```

```
Flags:
-h, --help help for oceanctl
```

Use "oceanctl [command] --help" for more information about a command.

- Step 4** Run the **kubectl get deploy -n  $\{NAMESPACE\}$**  command to check whether a Pod is started properly. In the preceding command,  $\{NAMESPACE\}$  indicates the namespace for installing CSM. **huawei-csm** is used as an example.

```
$ kubectl get deploy -n huawei-csm
NAME READY UP-TO-DATE AVAILABLE AGE
csm-prometheus-service 1/1 1 1 3h23m
csm-storage-service 1/1 1 1 3h23m
```

----End

## 8.1.2 Collecting CSM Logs Using oceanctl

- Step 1** Use a remote access tool, such as PuTTY, to log in to the node checked in [8.1.1 Performing Check Before Collection](#) through the management IP address.
- Step 2** Run the **oceanctl collect logs -n huawei-csm -a** command to collect CSM logs of all nodes where CSM containers reside in the cluster.

```
$ oceanctl collect logs -n huawei-csm -a
node[node2] Collection Progress: [+++++] 1/1 Pods
node[node3] Collection Progress: [+++++] 1/1 Pods
+++++
```

- Step 3** Check the log package generated in the **/tmp** directory. You can run the **unzip  $\{zip\_name\}$  -d collect\_logs** command to decompress the log package. In the preceding command,  $\{zip\_name\}$  indicates the package name.

```
$ date
Fri Sep 8 22:14:57 CST 2023

$ ls /tmp
huawei-csm-2023-09-08-22:13:01-all.zip
```

----End

## 8.2 Viewing Version Information

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the **kubectl get cm -n  $\{NAMESPACE\}$  huawei-csm-version -o yaml** command. In the preceding command,  $\{NAMESPACE\}$  indicates the namespace name, for example, **huawei-csm**.

```
$ kubectl get cm -n huawei-csm huawei-csm-version -o yaml
apiVersion: v1
data:
 cmi-controller: 2.7.0
 liveness-probe: 2.7.0
 prometheus-collector: 2.7.0
 topo-service: 2.7.0
kind: ConfigMap
metadata:
 name: huawei-csm-version
 namespace: huawei-csm
```

----End

# 9 Appendix

- [9.1 Accessing Kubernetes As a Non-root User](#)
- [9.2 Configuring the HTTPS Service for csm-prometheus](#)
- [9.3 Managing Storage Certificates](#)
- [9.4 Permission Matrix](#)

## 9.1 Accessing Kubernetes As a Non-root User

### Prerequisites

The non-root user has the sudo permission to use **/bin/cp** and **/bin/chown**.

### Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the **mkdir -p \$HOME/.kube** command to create a directory for storing the authentication file of the Kubernetes cluster.
- ```
$ mkdir -p $HOME/.kube
```
- Step 3** Run the **sudo cp -i /etc/kubernetes/admin.conf \$HOME/.kube/config** command to copy the authentication file of the Kubernetes cluster.
- Replace **/etc/kubernetes/admin.conf** with the actual authentication file.
- ```
$ sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
```
- Step 4** Run the **sudo chown \$(id -u):\$(id -g) \$HOME/.kube/config** command to change the user and user group of the authentication file.
- ```
$ sudo chown $(id -u):$(id -g) $HOME/.kube/config
```
- Step 5** Run the following commands to configure the **KUBECONFIG** environment variable for the current user (Ubuntu 20.04 is used as an example).
- ```
$ echo "export KUBECONFIG=$HOME/.kube/config" >> ~/.bashrc
$ source ~/.bashrc
```

----End

## 9.2 Configuring the HTTPS Service for csm-prometheus

### 9.2.1 Configuring the HTTPS Service Certificate of csm-prometheus

#### Prerequisites

- Kubernetes is running properly.
- You have obtained the certificate and key.

#### Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to the node where the **helm** folder of CSM is stored in the Kubernetes cluster through the management IP address.

**Step 2** Run the **cd /opt/huawei-csm/helm/huawei-csm** command to go to the Helm working directory.

**Step 3** Run the **mkdir cert** command to create a folder for storing the certificate.

**Step 4** Save the certificate and key files to the **/opt/huawei-csm/helm/huawei-csm/cert** directory.

```
-rw-r--r-- 1 root root 4.5K Sep 8 23:00 server.crt
-rw-r--r-- 1 root root 1.7K Sep 8 23:00 server.key
```

**Step 5** Run the **vim /opt/huawei-csm/helm/huawei-csm/values.yaml** command. Then:

- Set **features.prometheusCollector.prometheusCollectorSSL.enabled** to **true**.
- Set **features.prometheusCollector.prometheusCollectorSSL.certPath** to **"cert/server.crt"**.
- Set **features.prometheusCollector.prometheusCollectorSSL.keyPath** to **"cert/server.key"**.

```
all supported features
features:
 # prometheusCollector: allowed prometheus use the storage to collect metrics
 prometheusCollector:
 # Allowed values:
 # true: enable prometheus collect feature
 # false: disable prometheus collect feature
 # Default value: false
 enabled: true
 # nodePort: port the containers are provided to the prometheus
 # Default value: 30074
 nodePort: 30074
 # prometheusCollectorSSL: parameters required to start https
 # Default value: 30074
 prometheusCollectorSSL:
 # Allowed values:
 # true: enable https, when set it certPath and keyPath must set
 # false: disable https, use http
 # Default value: true
 enabled: true
 # The Path of cert, need to be placed in the huawei-csm directory
 certPath: "cert/server.crt"
```

```
The Path of key, need to be placed in the huawei-csm directory
keyPath: "cert/server.key"
```

**Step 6** Complete the installation by referring to [4.1.1 Installing the Software](#).

----End

## 9.2.2 Deleting the HTTPS Service Certificate of csm-prometheus

### Prerequisites

- Kubernetes is running properly.
- You have obtained the certificate and key.

### Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to the node where the **helm** folder of CSM is stored in the Kubernetes cluster through the management IP address.

**Step 2** Run the **cd /opt/huawei-csm/helm/huawei-csm** command to go to the Helm working directory.

**Step 3** Run the **helm get values huawei-csm -n huawei-csm -a > update-value.yaml** command to obtain the original service configuration file.

**Step 4** Run the **vim /opt/huawei-csm/helm/huawei-csm/update-value.yaml** command. Then:

- Set **features.prometheusCollector.prometheusCollectorSSL.enabled** to **false**.
- Set **features.prometheusCollector.prometheusCollectorSSL.certPath** to **""**.
- Set **features.prometheusCollector.prometheusCollectorSSL.keyPath** to **""**.

```
features:
 prometheusCollector:
 csiDriverName: csi.huawei.com
 enabled: true
 nodePort: 30074
 prometheusCollectorSSL:
 enabled: false
 certPath: ""
 keyPath: ""
```

**Step 5** Run the **helm upgrade huawei-csm ./ -n huawei-csm -f ./values.yaml -f update-value.yaml --wait --timeout 2m** command to upgrade CSM services. If **Release "huawei-csm" has been upgraded** is displayed in the command output, the CSM services are successfully upgraded.

----End

## 9.2.3 Updating the HTTPS Service Certificate of csm-prometheus

**Step 1** Use a remote access tool, such as PuTTY, to log in to the node where the **helm** folder of CSM is stored in the Kubernetes cluster through the management IP address.

**Step 2** Run the `cd /opt/huawei-csm/helm/huawei-csm` command to go to the Helm working directory.

**Step 3** Run the `mkdir cert` command to create a folder for storing the certificate.

**Step 4** Save the new certificate and key files to the `/opt/huawei-csm/helm/huawei-csm/cert` directory.

```
-rw-r--r-- 1 root root 4.5K Sep 8 23:00 server.crt
-rw-r--r-- 1 root root 1.7K Sep 8 23:00 server.key
```

**Step 5** Run the `helm get values huawei-csm -n huawei-csm -a > update-value.yaml` command to obtain the original service configuration file.

**Step 6** Run the `vim /opt/huawei-csm/helm/huawei-csm/update-value.yaml` command. Then:

- Set `features.prometheusCollector.prometheusCollectorSSL.enabled` to `true`.
- Set `features.prometheusCollector.prometheusCollectorSSL.certPath` to `"cert/server.crt"`.
- Set `features.prometheusCollector.prometheusCollectorSSL.keyPath` to `"cert/server.key"`.

```
features:
 prometheusCollector:
 csiDriverName: csi.huawei.com
 enabled: true
 nodePort: 30074
 prometheusCollectorSSL:
 enabled: true
 certPath: "cert/server.crt"
 keyPath: "cert/server.key"
```

**Step 7** Run the `helm upgrade huawei-csm ./ -n huawei-csm -f ./values.yaml -f update-value.yaml --wait --timeout 2m` command to upgrade CSM services. If **Release "huawei-csm" has been upgraded** is displayed in the command output, the CSM services are successfully upgraded.

----End

## 9.3 Managing Storage Certificates

For details, see section "Storage Backend Management" in the user guide of Huawei Storage Kubernetes CSI.

## 9.4 Permission Matrix

**Table 9-1** Permission matrix

| ClusterRole               | ApiGroups | Resources                                              | Verbs         |
|---------------------------|-----------|--------------------------------------------------------|---------------|
| prometheus-collector-role | -         | "persistentvolumes", "persistentvolume-claims", "pods" | "get", "list" |

| ClusterRole         | ApiGroups             | Resources                                         | Verbs                                                |
|---------------------|-----------------------|---------------------------------------------------|------------------------------------------------------|
|                     | "xuanwu.huawei.io"    | "storagebackendclaims"                            | "get", "list"                                        |
| cmi-collector-role  | "xuanwu.huawei.io"    | "storagebackendclaims"                            | "get"                                                |
|                     | -                     | "secrets"                                         | "get"                                                |
|                     | -                     | "configmaps"                                      | "create", "get", "update"                            |
| topo-service-role   | -                     | "secrets", "events", "configmaps"                 | "create", "get", "update", "delete"                  |
|                     | "coordination.k8s.io" | "leases"                                          | "create", "get", "update", "delete"                  |
|                     | "xuanwu.huawei.io"    | "resourcetopologies", "resourcetopologies/status" | "create", "get", "list", "watch", "update", "delete" |
|                     | "*"                   | "*"                                               | "get", "list", "watch"                               |
| cmi-controller-role | "xuanwu.huawei.io"    | "storagebackendclaims"                            | "get"                                                |
|                     | -                     | "secrets"                                         | "get"                                                |
|                     | -                     | "configmaps"                                      | "create", "get", "update"                            |



# 10 FAQs

10.1 Pod in OOMKilled State

10.2 Manually Adjusting the Number of Concurrent Data Scraping Requests

10.3 Residual Pod Labels on the Storage Side

10.4 The Pod Status Is CrashLoopBackOff and the Log Contains "mkdir permission denied"

## 10.1 Pod in OOMKilled State

### Symptom

The Pod state is **OOMKilled**, and alarm "unable to set memory limit to xxx (current usage: xxx, peak usage: xxx): unknown" is generated in the Pod.

```
[root@k8s-master1-yqz CSMImage]# kubectl get pod -n huawei-csm csm-prometheus-service-65b446d597-ljx7q
NAME READY STATUS RESTARTS AGE
csm-prometheus-service-65b446d597-ljx7q 2/3 OOMKilled 3 (<invalid> ago) 43s
```

```
Warning Failed <invalid> kubelet Error: failed to create containerd task: failed to create shim task: OCI runtime create failed: runc create failed: unable to start container process: error during container init: error setting cgroup config for procHooks process: unable to set memory limit to 5242880 (current usage: 5631804, peak usage: 7139326): unknown
```

### Root Cause Analysis

The container memory usage in the Pod reaches the upper limit. You need to increase the upper limit of the available memory.

### Solution

Modify the container resource configuration and update CSM. For details, see [4.1.3 Upgrading the Software](#).

## 10.2 Manually Adjusting the Number of Concurrent Data Scraping Requests

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the **kubectrl edit deployments.apps -n huawei-csm csm-prometheus-service** command to edit the **deployments** configuration item.

**Step 3** For example, to set the number of concurrent requests to 10, add **args** list item **--client-max-threads=10** to the container whose **name** is **cmi-controller** as follows.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: csm-prometheus-service
 namespace: huawei-csm
...
spec:
 template:
 metadata:
 creationTimestamp: null
 labels:
 app: csm-prometheus-service
 spec:
 containers:
...
 - args:
 - --cmi-address=$(ENDPOINT)
 - --cmi-name=cmi.huawei.com
 - --page-size=100
 - --backend-namespace=huawei-csi
 - --log-file-dir=/var/log/huawei-csm/csm-prometheus-service
 - --log-file=cmi-service
 - --logging-module=file
 - --log-level=info
 - --log-file-size=20M
 - --max-backups=9
 - --client-max-threads=10
 name: cmi-controller
...
...
```

**Step 4** Run the **:wq** command to save the configuration and exit.

**Step 5** Run the **kubectrl get pod -n huawei-csm** command and wait until the container is restarted.

**Step 6** Run the **kubectrl get deployments.apps -n huawei-csm csm-prometheus-service -o yaml | grep client-max-threads** command to check whether the configuration is successful.

----End

## 10.3 Residual Pod Labels on the Storage Side

### Symptom

After CSM is upgraded from 1.x.x to 2.x.x, residual Pod topo relationships exist on storage resources and cannot be cleared.

### Root Cause Analysis

In 1.x.x, the name of a topo resource is the name of a storage resource. In 2.x.x, the name of a topo resource is changed to a PV name in the cluster, and the prefix is changed from **topo-** to **rt-**.

### 1.x.x:

| NAME                                          | PROVISIONER    | VOLUMEHANDLE                                       |
|-----------------------------------------------|----------------|----------------------------------------------------|
| STATUS AGE                                    |                |                                                    |
| topo-pvc-0a8f7871-f26e-4665-b4be-53dfa9c878cb | cmi.huawei.com | 181-iscsi.pvc-0a8f7871-f26e-4665-b4be-53dfa9c878cb |
| Normal 353d                                   |                |                                                    |
| topo-pvc-3b6b1dd6-b3dd-4394-8251-4bd6009411cf | cmi.huawei.com | 181-iscsi.pvc-3b6b1dd6-b3dd-4394-8251-4bd6009411cf |
| Normal 354d                                   |                |                                                    |

### 2.x.x and later:

| NAME                                        | PROVISIONER    | VOLUMEHANDLE                                       |
|---------------------------------------------|----------------|----------------------------------------------------|
| STATUS AGE                                  |                |                                                    |
| rt-pvc-0a8f7871-f26e-4665-b4be-53dfa9c878cb | cmi.huawei.com | 181-iscsi.pvc-0a8f7871-f26e-4665-b4be-53dfa9c878cb |
| Normal 353d                                 |                |                                                    |
| rt-pvc-3b6b1dd6-b3dd-4394-8251-4bd6009411cf | cmi.huawei.com | 181-iscsi.pvc-3b6b1dd6-b3dd-4394-8251-4bd6009411cf |
| Normal 354d                                 |                |                                                    |

After CSM is upgraded to 2.x.x, the topology service creates new topo resources based on the resource information in the cluster and binds the new topo resources to the topology relationships of the source version on the storage side.

If a Pod whose topology relationship has been to the storage device is deleted when the CSM topology service is disabled, the old topology relationship bound to the topo resource cannot be created based on the existing resources in the cluster after the CSM topology service is enabled again.

In this case, the old topology relationship remains on the storage device and cannot be cleared.

## Solution or Workaround

If the preceding storage topology relationship remains, you can only delete the storage resources to clear the residual topology relationship.

During the upgrade from CSM 1.x.x to 2.x.x, if the CSM topology service is enabled in the source version, enabling the CSM topology service during the upgrade can prevent the preceding problem.

## 10.4 The Pod Status Is CrashLoopBackOff and the Log Contains "mkdir permission denied"

### Symptom

On the OpenShift platform, the log mode is **file**. During CSM installation, the Pod status is **CrashLoopBackOff** and the following error information is displayed in the log:

```
init log error: [could not initialize logging to file: could not create log directory /var/log/huawei-csm/csm-prometheus-service. mkdir /var/log/huawei-csm: permission denied]
```

### Root Cause Analysis

Due to OpenShift platform restrictions or user security configurations, the CSM container does not have sufficient permission to create log directories on the host.

## Solution or Workaround

Solution 1: Change the log mode to **console**.

Solution 2: Manually plan the CSM log directory on the corresponding nodes. The procedure is as follows:

- Step 1** Use a remote access tool, such as PuTTY, to log in to a node in the Kubernetes cluster through the management IP address.
- Step 2** If the container platform is Kubernetes, run the **mkdir -p /var/log/huawei-csm && chmod 757 /var/log/huawei-csm** command to create a log directory and set the DAC permission of the log directory to **757**.

```
mkdir -p /var/log/huawei-csm && chmod 757 /var/log/huawei-csm
```

If the container platform is OpenShift, run the **mkdir -p /var/log/huawei-csm && chmod 757 /var/log/huawei-csm && chcon -t svirt\_sandbox\_file\_t /var/log/huawei-csm** command to create a log directory, and set the DAC permission of the log directory to **757** and the SELinux permission to **svirt\_sandbox\_file\_t**.

```
mkdir -p /var/log/huawei-csm && chmod 757 /var/log/huawei-csm && chcon -t
svirt_sandbox_file_t /var/log/huawei-csm
```

- Step 3** Repeat the preceding steps to plan the **/var/log/huawei-csm** log directory on the nodes where Huawei CSM container runs.

---

### NOTICE

Ensure that the **/var/log/huawei-csm** log directory has been planned for all nodes that may be scheduled by Huawei CSM container. If node failover occurs during the running of Huawei CSM container and the log directory is not planned for the new node where the container runs in advance, the container cannot be started due to insufficient permission.

---

----End